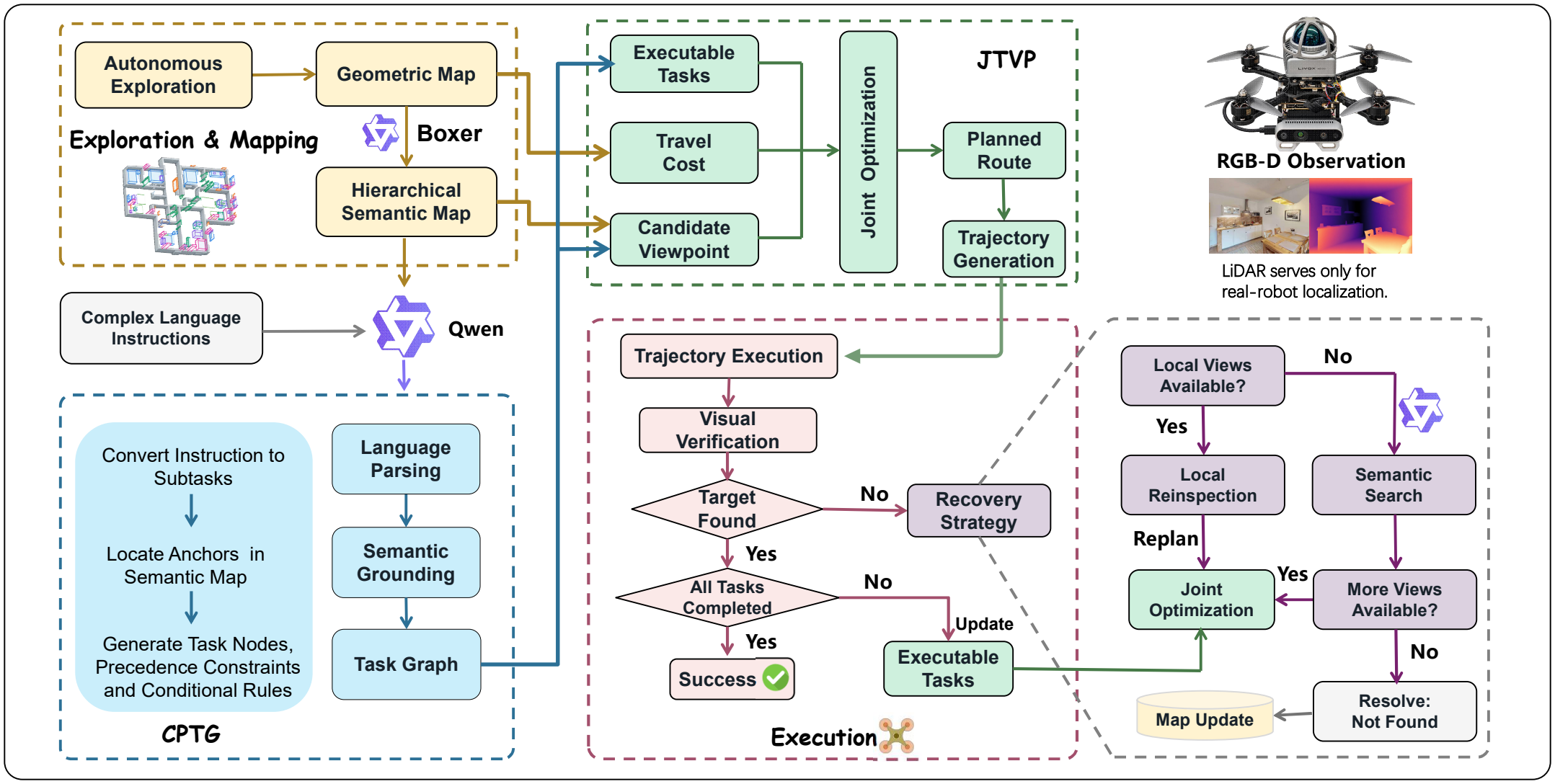

Autonomous Exploration
Geometric Map
Exploration & Mapping
Boxer
Hierarchical Semantic Map
Complex Language Instructions
Qwen
Convert Instruction to Subtasks
Locate Anchors in Semantic Map
Generate Task Nodes, Precedence Constraints and Conditional Rules
Language Parsing
Semantic Grounding
Task Graph
CPTG
Executable Tasks
Travel Cost
Candidate Viewpoint
Joint Optimization
JTVP
Planned Route
Trajectory Generation
Trajectory Execution
Visual Verification
Target Found
No
Recovery Strategy
Yes
All Tasks Completed
No
Update
Yes
Success
Executable Tasks
Execution
Local Views Available?
No
Yes
Local Reinspection
Semantic Search
Replan
Joint Optimization
Yes
More Views Available?
No
Map Update
Resolve: Not Found
RGB-D Observation
LiDAR serves only for real-robot localization.